

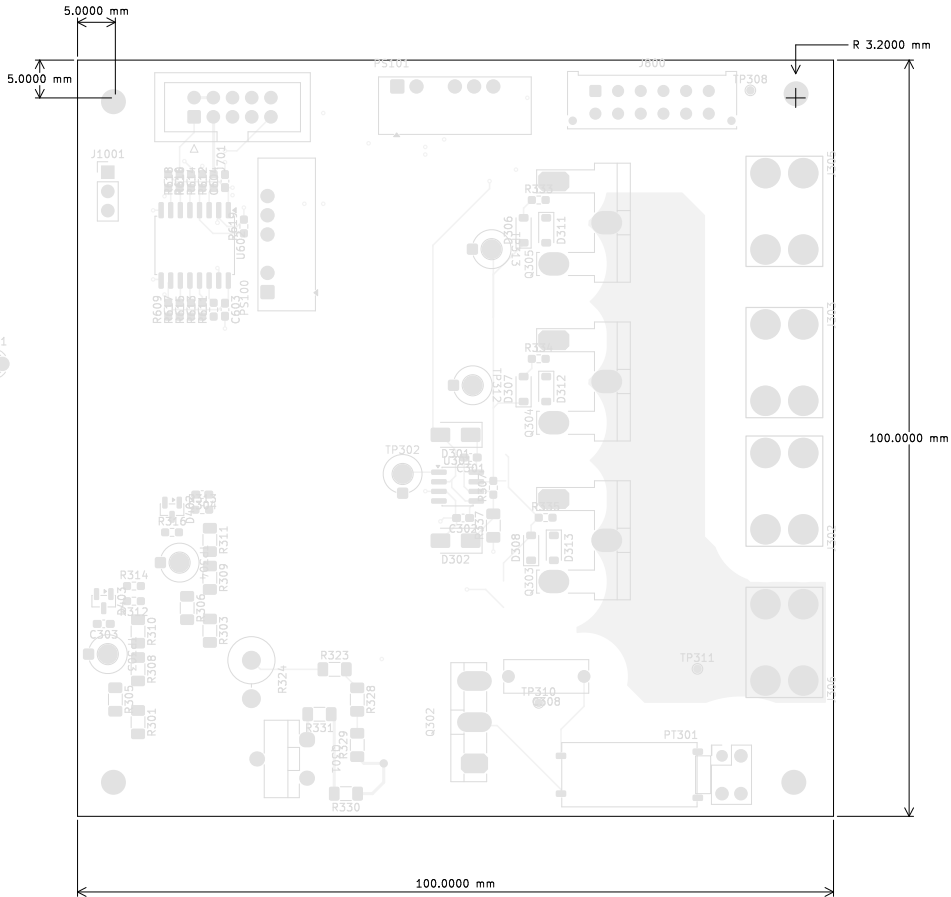
Main board Fabrication Document

Top Fabrication (Scale 1:1)

Layer Stack Legend

| Material     | Layer             | Thickness        | Dielectric    | Type        | Gerber |
|--------------|-------------------|------------------|---------------|-------------|--------|
| F.Paste      |                   |                  |               | Paste Mask  |        |
| F.Silkscreen |                   |                  |               | Legend      | GBR    |
| F.Mask       |                   | 0.02mm           | Solder Resist | Solder Mask | GBR    |
| Copper       | F.Cu (Sig, PWR)   | 0.07mm (2.00oz)  |               | Signal      | GBR    |
| Prepreg      |                   | 0.1mm            | FR4_7628      | Dielectric  |        |
| Copper       | In1.Cu (GND)      | 0.035mm (1.00oz) |               | Plane       | GBR    |
| Core         |                   | 1.828mm          | FR4           | Dielectric  |        |
| Copper       | In2.Cu (Sig, PWR) | 0.035mm (1.00oz) |               | Plane       | GBR    |
| Prepreg      |                   | 0.1mm            | FR4_7628      | Dielectric  |        |
| Copper       | B.Cu (Sig, PWR)   | 0.07mm (2.00oz)  |               | Signal      | GBR    |
| B.Mask       |                   | 0.02mm           | Solder Resist | Solder Mask | GBR    |
| B.Silkscreen |                   |                  |               | Legend      | GBR    |
| B.Paste      |                   |                  |               | Paste Mask  |        |

Total thickness: 2.278mm  
Note: external layer thicknesses are specified after plating



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- FABRICATE PER IPC-6012A CLASS 2.
- OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge\_Cuts,GBR" SUFFIX.  
  
DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.  
  
SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- SURFACE FINISH: HAL SNPB
- SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
- SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- PCB MATERIAL REQUIREMENTS:
  - FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
  - Tg 170 C OR EQUIVALENT.
  - EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY CHURRASCOD ELECTRONICS.
- DESIGN GEOMETRY MINIMUM FEATURE SIZES:

|                  |                      |
|------------------|----------------------|
| BOARD SIZE       | 100.000 × 100.000 mm |
| BOARD THICKNESS  | 2.278 mm             |
| TRACE WIDTH      | 0.200 mm             |
| TRACE TO TRACE   | 0.200 mm             |
| MIN. HOLE (PTH)  | 0.300 mm             |
| MIN. HOLE (NPTH) | 1.000 mm             |
| ANNULAR RING     | 0.150 mm             |
| COPPER TO HOLE   | 0.300 mm             |
| COPPER TO EDGE   | 0.500 mm             |
| HOLE TO HOLE     | 0.250 mm             |

Lim. resistor  
PFC

HMI

Ext  
ctrl

SWD  
Status

1 kV strobe  
04/Jul/2025

Boot

Light  
sense

Felipe L Rodrigues

Cap+

Cap-

Lamp  
trig

Lamp+

Lamp-

|                                                                                                                    |              |            |                        |            |                     |           |
|--------------------------------------------------------------------------------------------------------------------|--------------|------------|------------------------|------------|---------------------|-----------|
| <br>ChurrasCod<br>Electronics | Comments:    |            | Company:               |            | Variant:            | Git Hash: |
|                                                                                                                    |              |            | ChurrasCod Electronics |            | PRELIMINARY         | f657b35   |
|                                                                                                                    | Sheet Title: |            | Board Name:            |            | Project Name:       |           |
|                                                                                                                    |              |            | Main board             |            | 1 kV strobe         |           |
| Top Fabrication (Scale 1:1)                                                                                        |              | File Name: | Designer:              | Date:      | Revision:           |           |
| Strobe_main.kicad_pcb                                                                                              |              | FR         |                        | 2024-04-13 | 1.0.0+ (Unreleased) |           |
| Sheet Path:                                                                                                        |              |            | Reviewer:              | Size:      | Sheet:              |           |
|                                                                                                                    |              |            |                        | A3         | 1 of 10             |           |